NAME:-Sheikh Wasimuddin

ROLL NO:-73

BRANCH :-CSE(IOT)

LAB 09	
Aim:	Introduction to Packages in R. Data visualization using package ggplot2
Theory	Data visualization is an essential aspect of data analysis, providing insights through graphical representation. R offers two major plotting systems: base R graphics and the ggplot2 package. While base R provides fundamental plotting capabilities, ggplot2 introduces an advanced, structured, and flexible approach to visualization, built upon the Grammar of Graphics framework. 1. Grammar of Graphics Framework ggplot2 is built on Wilkinson's Grammar of Graphics, which breaks down visualizations into a set of independent components: Data: The dataset to be visualized. Aesthetics (aes): The mapping of data variables to visual properties (e.g., color, shape, size). Geometries (geom): The type of plot (e.g., points, lines, bars). Facets: The arrangement of multiple plots based on a variable. Statistics: Transformations applied to data before visualization (e.g., smoothing, binning). Coordinates: The coordinate system used (e.g., Cartesian, polar). Themes: The overall appearance of the plot. This modular structure makes ggplot2 more systematic and scalable than base R graphics. 2. Improved Aesthetics and Customization ggplot2 automatically applies modern design principles to produce wellstructured plots. It provides built-in themes (theme_minimal (), theme_classic(), etc.), allowing easy customization. Base R, in contrast, requires manual adjustments using par (), axis (), and other low-level functions, making customization cumbersome. 3. Layered Approach to Plot Construction One of the biggest advantages of ggplot2 is its layering system. Instead of using separate functions for adding elements (as in base R), ggplot2 allows users to build complex visualizations by adding layers incrementally using the + operator. This facilitates better organization and easier modification of plots. For example, adding a regression line to a scatter plot in ggplot2 simply requires an additional geom_smooth() layer. In base R, the same task requires multiple function calls and careful handling of graphical parameters.
Program:	#sheikh wasimuddin 23071044 cse iot 73 # Load required package library(ggplot2) # Load built-in datasets data(mpg) data(mtcars)

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```
# 1   Scatter Plot (LOESS Smoothing Curve)
ggplot(mpg, aes(x = displ, y = hwy)) +
  geom_point() +
  geom_smooth(method = "loess", color = "blue") +
  ggtitle("LOESS Smoothing Curve")

# 2   Histogram Binning
ggplot(mpg, aes(x = hwy)) +
  geom_histogram(binwidth = 5, fill = "blue", color = "black") +
  ggtitle("Binning in Histogram")

# 3   Scatter Plot (Horsepower vs MPG)
ggplot(mtcars, aes(x = hp, y = mpg)) +
  geom_point(color = "blue", size = 3) +
  labs(title = "MPG vs Horsepower", x = "Horsepower", y = "Miles Per Gallon") +
  theme_minimal()

# 4   Bar Chart (Categorical Data - Cylinder Count)
ggplot(mtcars, aes(x = factor(cyl))) +
  geom_bar(fill = "purple") +
  labs(title = "Number of Cars by Cylinder Count", x = "Cylinders", y = "Count")

# 5   Histogram (Distribution of MPG)
ggplot(mtcars, aes(x = mpg)) +
  geom_histogram(fill = "green", color = "black", bins = 10) +
```

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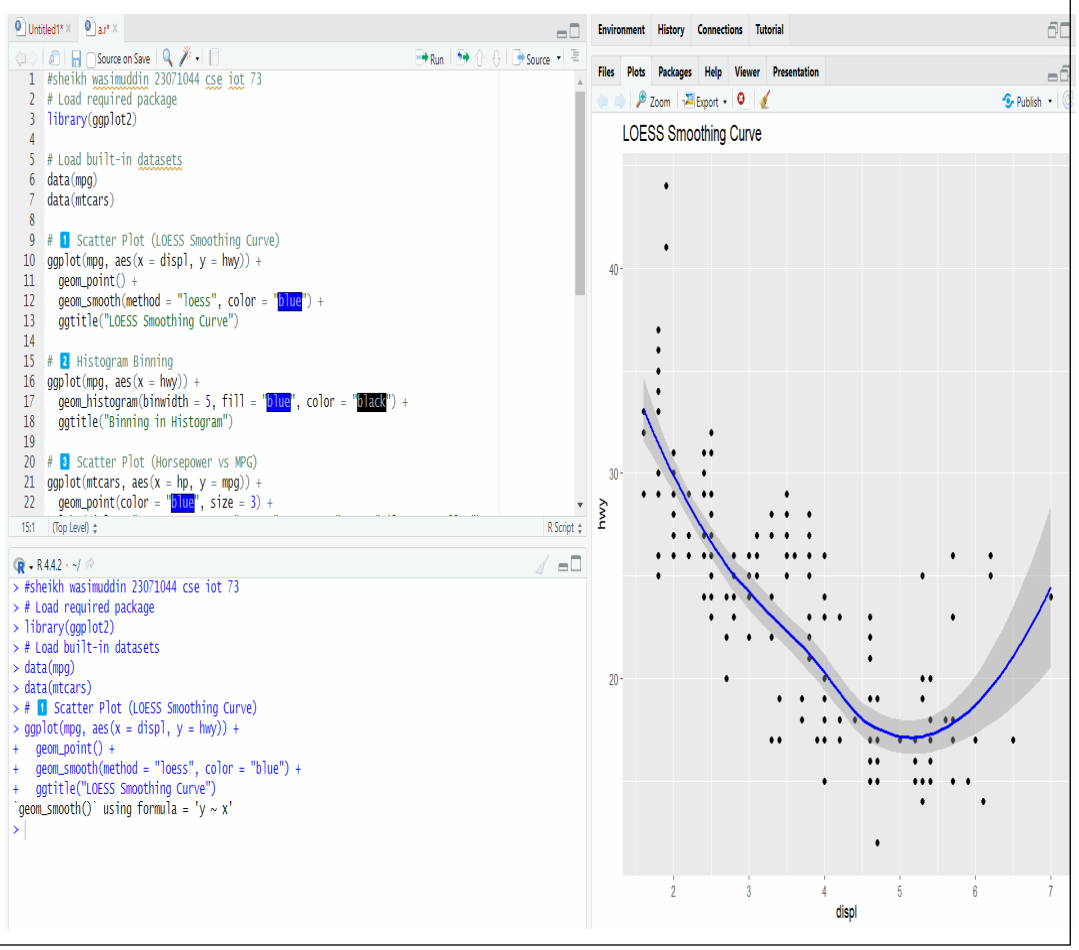
```
  labs(title = "Histogram of MPG", x = "Miles Per Gallon", y = "Frequency")


# 6   Boxplot (MPG by Cylinder Count)

ggplot(mtcars, aes(x = factor(cyl), y = mpg)) +

  geom_boxplot(fill = "orange") +

  labs(title = "MPG by Cylinder Count", x = "Cylinders", y = "Miles Per Gallon")
```

Output:



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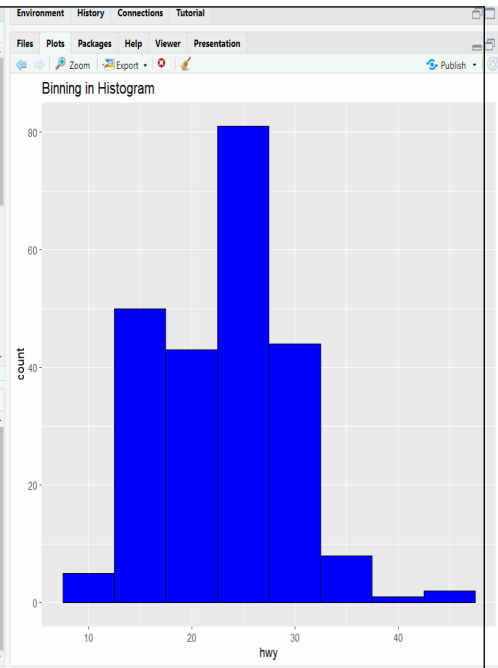


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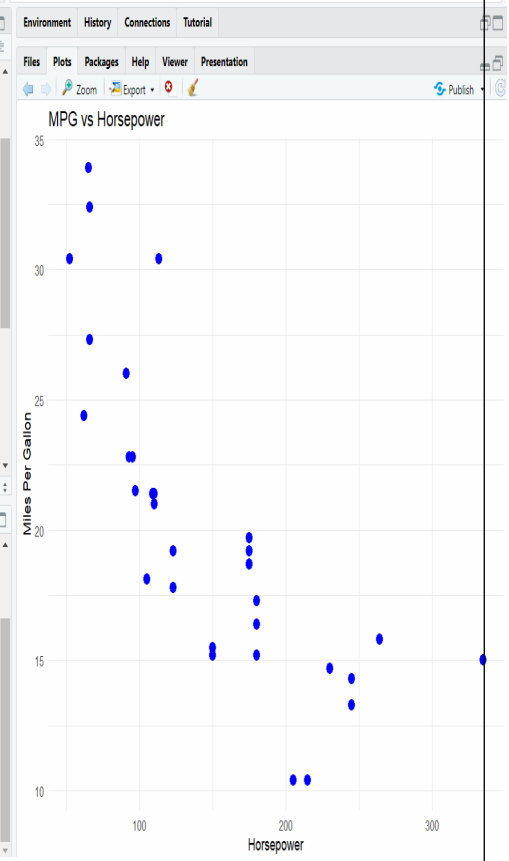

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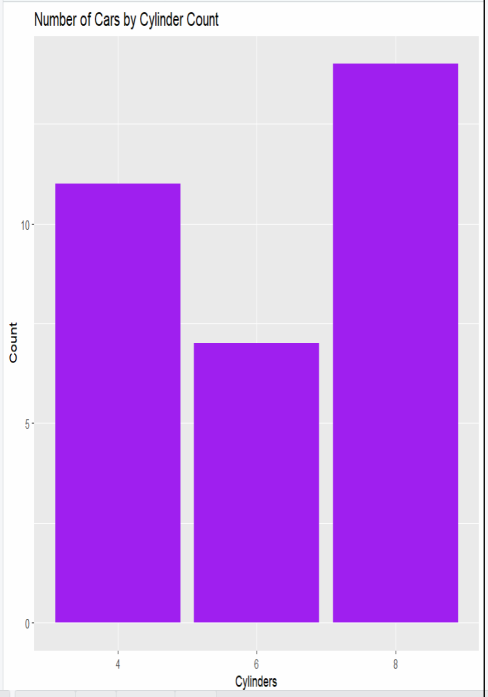


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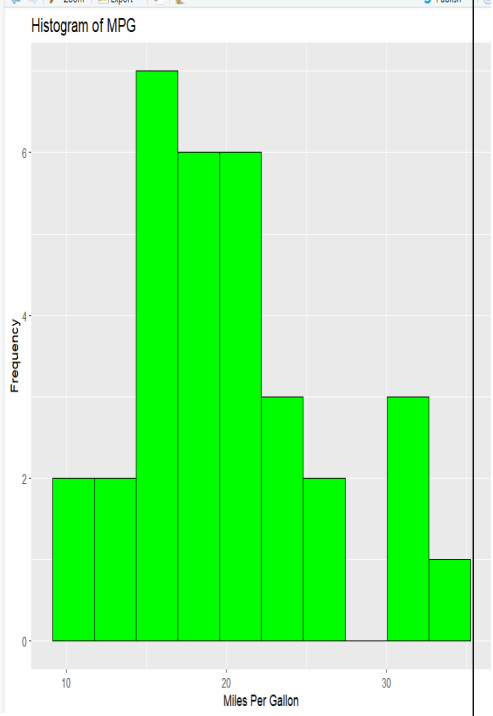

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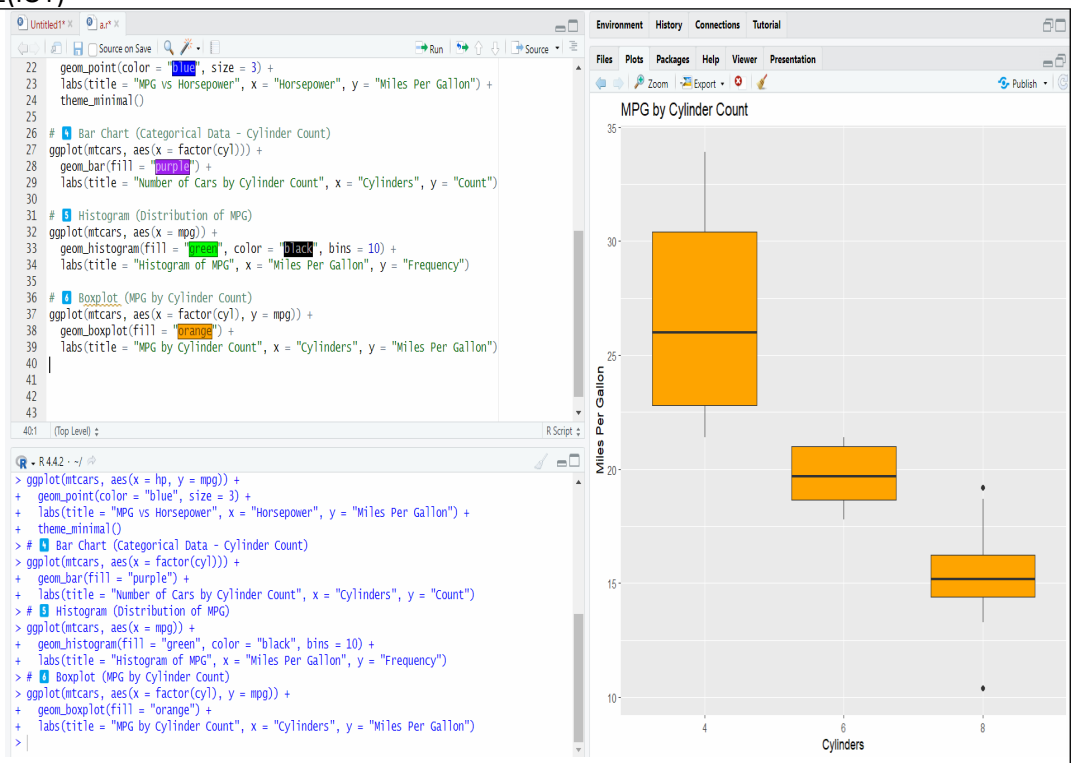

Conclusion	Introduction to Packages in R. Data visualization using package ggplot2 executed successfully
Name Of Faculty:	Prof. Akhil Jajulwar